

Alberto Berral-González

An enthusiast specialized in bioinformatics and computational biology, solving real-world problems using the combination of computer science, biology and statistics with passion for friendly code design, best practices and documentation.

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EXPERIENCE

IBMCC, Salamanca — Researcher (MOMENTUM)

DEC 2024 - PRESENT

- Updated and curated the APID Interactomes and GEDA databases, ensuring accurate protein-protein and drug-target interaction networks backed by scientific literature.
- Prepared the human interactome for integration into a deep neural network (DNN)-based large language model (LLM).
- Developed an LLM incorporating pharmacogenomic data, making it programmatically accessible for drug-target interaction analysis.
- Built predictive models using deep neural networks (DNN) to infer weighted protein relationships and analyze intercellular communication (scGPT, LIANA).
- **Tools:** R, Python

IBMCC, Salamanca — Researcher (IMPACT-DATA)

JUL 2022 - DEC 2024

- Implemented federated EGA instances for secure genomic data storage and management.
- Adapted and deployed open-source software for multi-source data analysis and integration.
- Integrated genomic, radiomic, and structured medical data for secondary use applications.
- Developed automated systems for identifying relevant genomic and medical imaging data, providing anonymized responses in a beacon-style format.
- **Tools:** R, Open-Source Software

ARTELNICS, Salamanca — Data Scientist (DIPCAN)

JAN 2022 - JUL 2022

- Integrated clinical, genomic, pathological, and radiomic data from 2,000 patients with metastatic non-hematologic tumors to advance personalized cancer treatment.
- Developed an AI-driven algorithm to enhance precision medicine by identifying patients more likely to respond to specific therapies.
- Improved digital health transformation in oncology, enabling systematic and efficient patient data collection and analysis.
- Optimized cost-effective strategies for cancer prevention, diagnosis, and treatment through interdisciplinary collaboration.
- **Tools:** R, Python

IBMCC, Salamanca — Researcher (ARREST-AD)

DEC 2018 - DEC 2021

- Investigated the role of heparan sulfate in Alzheimer's disease to develop novel diagnostic and therapeutic strategies.
 - Conducted genomic and proteomic analysis on patient samples to identify biomarkers for early detection.
 - Developed an AI-assisted approach to study biochemical mechanisms underlying neurodegeneration.
 - Enhanced personalized medicine strategies for Alzheimer's treatment through interdisciplinary collaboration.
 - **Tools:** R, Python
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EDUCATION

Universidad de Salamanca (USAL), Salamanca — PhD

PhD in Biosciences: biology and clinic of cancer and translational medicine

2019 - 2024

Universitat de València (UV), Valencia — MSc.Bioinformatics

Master in Bioinformatics

2015 - 2017

Universidad de Córdoba (UCO), Córdoba — B.Biology

Bachelors in Biology

2010 - 2024

PUBLICATIONS

2025

Novel Assignment of Gene Markers to Hematological and Immune Cells Based on Single-Cell Transcriptomics. De La Rosa, Enrique; Alonso-Moreda, Natalia; Berral-González, Alberto; Sánchez-Luis, Elena; González-Velasco, Oscar; Sánchez-Santos, José Manuel; De Las Rivas, Javier. *International Journal of Molecular Sciences*, Vol. 26, Núm. 2, p. 805.

2024

Stratification of colorectal patients based on survival analysis shows the value of consensus molecular subtypes and reveals the CBLL1 gene as a biomarker of CMS2 tumours. Alfonsín, Gloria; Berral-González, Alberto; Rodríguez-Alonso, Andrea; Quiroga, Macarena; De Las Rivas, Javier; Figueroa, Angélica. *International Journal of Molecular Sciences*, Vol. 25, Núm. 3, p. 1919.

RAS-p110α signalling in macrophages is required for effective inflammatory response and resolution of inflammation. Rosell, Alejandro; Krygowska, Agata A.; Pérez, Marta Alcón; Voisin, Mathieu-Benoit; de Paz, Juan; Sanz-Fraile, Héctor; Rajeeve, Vinothini; Berral-González, Alberto; Cuesta, Cristina; Swinyard, Otilie, et al. *eLife*, Vol. 13.

Cross-species regulatory network analysis identifies FOXO1 as a driver of ovarian follicular recruitment. Kramer, Ashley E.; Berral-González, Alberto; Ellwood, Kathryn M.; Ding, Shanshan; De Las Rivas, Javier; Dutta, Aditya. *Scientific Reports*, Vol. 14, Núm. 1, p. 30787.

2023

Identification of a gene expression signature associated with breast cancer survival and risk that improves clinical genomic platforms. Bueno-Fortes, Santiago; Berral-González, Alberto; Sánchez-Santos, José Manuel; Martín-Merino, Manuel; De Las Rivas, Javier. *Bioinformatics Advances*, Vol. 3, Núm. 1, p. vbad037.

Comparative analysis of cell mixtures deconvolution and gene signatures generated for blood, immune and cancer cells. Alonso-Moreda, Natalia; Berral-González, Alberto; De La Rosa, Enrique; González-Velasco, Oscar; Sánchez-Santos, José Manuel; De Las Rivas, Javier. *International Journal of Molecular Sciences*, Vol. 24, Núm. 13, p. 10765.

2021

Cancer-associated fibroblast-derived gene signatures determine prognosis in colon cancer patients. Herrera, Mercedes; Berral-González, Alberto; López-Cade, Igor; Galindo-Pumariño, Cristina; Bueno-Fortes, Santiago; Martín-Merino, Manuel; Carrato, Alfredo; Ocaña, Alberto; De La Pinta, Carolina; López-Alfonso, Ana, et al. *Molecular Cancer*, Vol. 20, Núm. 1, p. 73.

A gene signature derived from the loss of CDKN1A (p21) is associated with CMS4 colorectal cancer. Bueno-Fortes, Santiago; Muenzner, Julianne K.; Berral-González, Alberto; Hampel, Chuanpit; Lindner, Pablo; Berninger, Alexandra; Huebner, Kerstin; Kunze, Philipp; Bäuerle, Tobias; Erlenbach-Wuenssch, Katharina, et al. *Cancers*, Vol. 14, Núm. 1, p. 136.

2020

Mining drug-target associations in cancer: analysis of gene expression and drug activity correlations. Arroyo, Monica M.; Berral-González, Alberto; Bueno-Fortes, Santiago; Alonso-López, Diego; De Las Rivas, Javier. *Biomolecules*, Vol. 10, Núm. 5, p. 667.

2019

OMICfpp: a fuzzy approach for paired RNA-Seq counts. Berral-González, Alberto; Riffo-Campos, Angela L.; Ayala, Guillermo. *BMC Genomics*, Vol. 20, p. 1-20.